

Emiliano Contreras Rivera

Mechatronics Engineering Student

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Work Experience

Participant

E3 Semiconductors UNISON Program | 2025

- Learned about the semiconductor industry and its applications.
- Strengthened ethical awareness and social responsibility.

Semiconductor fundamentals Industry practices CSR

Trainee

Santander Agile Program | 2024

- Learned agile methodology principles.
- Worked in teams following agile workflows.

Agile Team collaboration

Education

Mechatronics Engineering Degree

Universidad de Sonora | Expected May 2028

International Exchange

Politechnika Łódzka, Poland | 2026-2

Projects

Adjustable DC Power Supply: Hands-on design and implementation of an adjustable DC power supply.

Wave rectifier Power Circuits Filter π

SCARA Robotic Arm: Design and construction of a SCARA-type robotic arm prototype.

Mechanisms Robotic Control

Humidity and Temperature Monitoring System: Hands-on design and implementation of the system, featuring network communication between two microcontrollers and database access via Telegram.

Docker Node-Red MySQL Chat-bot

Skills

- Excellent communication and teamwork skills.
- Strong sense of responsibility and kindness.
- Motivated to encourage group performance.
- Ability to learn quickly and adapt.

Technical Skills

Manufacturing Processes: Welding, turning, milling machine, temporary mold casting, polymers.

Electronics: Electronic symbols, low-pass and high-pass filters, amplifiers, logic gates, flip-flops, wave rectifiers.

Programming: Python, C, Java.

Mechanics: Design of mechanical elements and mechanisms.

Control: Basic knowledge of block diagrams and control systems.

Software: Python, Google Colab, NetBeans, Livewire, SolidWorks, Docker, Arduino, FluidSIM.

Projects

Prototypes and Competitions

2023 - Present

Hands-on learning through practical prototypes and participation in engineering competitions, including the 3rd MAKERSPACE EXPO.

Prototyping Functionality Electrical Design

Up-to-date version of resume is available at 0xemiss.github.io